

Applied Mathematics – 1st Periodical Examination (September 2024)

[Note: the college name, course code and top header were cut off in the original photo]

Branch: CE

Course Advisor: **Bit Rana**

Question Paper for 1st Periodical Examination (Sep 2024)

Time: 2 Hours

MaxMarks: 50

Note: Attempt All Question

At the end of this course students will demonstrate the ability to:

- 1) Examine linear simultaneous equations for consistency and applications
- 2) Apply concept of linear algebra to convert quadratic form to canonical forms
- 3) Solve problems on improper integral & Inspect the function of two variables for optimization.
- 4) Evaluate double and triple integrals in various coordinates & their application in engineering
- 5) Test for the analyticity of complex functions.
- 6) Solve problems on complex integration.

Q1(co1)(a) Show that $A =$

$$A = \begin{vmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{vmatrix} \text{ is orthogonal.}$$

(b) Every square matrix can be uniquely expressed as a sum of Hermitian and skew-Hermitian Matrices.

(c) Find the rank of the given matrix $A =$

$$A = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 4 & 4 \\ 7 & 10 & 12 \end{vmatrix}$$

(d) Show that the vectors $(1, 2, 3)$, $(3, -2, 1)$, $(1, -6, -5)$ form a linearly dependent set.

(e) State the Rolle's theorem and Lagrange's mean value theorem

(2×5=10)

Q2(co1) Find the Eigen values and the corresponding Eigen vectors of the matrix.

$$\begin{vmatrix} -2 & 2 & 3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{vmatrix}$$

(8×5=40)

Q3(co1) Determine the modal matrix P of $A =$

$$\begin{vmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{vmatrix}$$

Verify that $P^{-1}AP$ is a diagonal matrix.

Q4(co1) Solve following system of equations $x+2y+2z=2$, $3x-2y+z=5$, $2x-5y+3z=-4$, $x+4y+6z=0$ with Gauss elimination method

Q5.(co1) Using Cayley - Hamilton Theorem find the inverse and A

$$A = \begin{vmatrix} 7 & 2 & -2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{vmatrix}$$

⁴ of the matrix A shown above.

Q6(co2) Reduction quadratic to canonical form $3x^2+5y^2+3z^2-2yz+2zx-2xy$ specify matrix transformation